



$$\delta = \psi - \phi = \text{---}^\circ$$

$$E_f = V \cos \delta + IR_a \cos \psi - IX_d \sin \psi = \text{---} V$$

$$\% \text{ regulation} = \frac{E_f - V}{V} \times 100 \% = \text{---} \%$$

% regulation at half load $V = \text{---} V$, $I = \text{---} A$

Power factor	ϕ	ψ	δ	E_f	Regulation
0 lag					
0.2 lag					
0.4 lag					
0.6 lag					
0.8 lag					
1					
0.8 lead					
0.6 lead					
0.4 lead					
0.2 lead					
0 lead					

MODEL GRAPH

